

KUNDAN KUMAR PANDEY

Senior Data Engineer · Distributed Systems · Real-Time Streaming · Lakehouse · Cloud-Native

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PROFESSIONAL SUMMARY

Senior Data Engineer with 6 years designing and operating large-scale distributed data platforms in production. Architected streaming infrastructure for Marriott International (500+ hotels, 10K+ events/sec, sub-5s SLA) and contract intelligence platform for Morgan Stanley (2M+ docs/month). Expert in Spark internals, Kafka, Delta Lake, and AWS. Comfortable owning system design end-to-end — trade-off analysis, fault-tolerance strategy, and cross-functional alignment. Delivered 40% Spark performance gains, 24x SLA improvement, and 60% QA reduction in production.

SYSTEM DESIGN HIGHLIGHTS

- Designed a real-time distributed pipeline processing 10K+ events/sec across 500+ global properties with exactly-once guarantees, stateful stream processing, and 99.9% SLA — using Kafka + Spark Structured Streaming with transactional Delta Lake writes.
- Architected fault-tolerant ingestion at 2M+ docs/month with backpressure handling, dead-letter queue strategy, circuit-breaker failover, and automatic checkpointing — zero message loss under upstream API failures.
- Led Kappa vs. Lambda architecture decision for a production streaming platform — trade-off analysis, stakeholder alignment, and full design doc ownership — eliminating dual-codebase complexity.

TECHNICAL SKILLS

Languages: Python, PySpark, Scala, Java, SQL, Groovy

Streaming & Batch: Apache Spark (Batch & Streaming), Kafka, NiFi, Flink (familiar), Stateful Stream Processing, Checkpointing Strategy

Cloud & Infra: AWS (S3, EMR, Redshift, Lambda, Glue, Kinesis), Databricks, GCP/BigQuery (familiar), Terraform (familiar), Autoscaling

Data Architecture: Lakehouse (Delta Lake), Kappa/Lambda Architecture, ETL/ELT, SCD Type 2, Exactly-Once Semantics, Data Contracts

Spark Tuning: AQE, Partition Pruning, Broadcast Joins, Skew Salting, Z-Ordering, JVM Tuning, Memory Tuning, Dynamic Partition Coalescing

Reliability: Fault-Tolerance Strategy, Backpressure Handling, DLQ, Circuit Breaker, On-Call Runbooks, MTTD/MTTR Optimization

Tooling: Airflow, Oozie, Jenkins, Git, CI/CD, dbt (familiar), Avro, Schema Registry, AWS Glue Catalog, Great Expectations (familiar)

WORK EXPERIENCE

Senior Data Engineer · LTIMindtree (Client: Marriott International) · Bengaluru, India *Jan 2025 – Present*

- Authored design doc and led architecture decision to adopt Kappa over Lambda — eliminated dual-codebase maintenance, reduced operational complexity by 35%; system processes 10M+ records/day at 99.9% uptime.
- Cut Spark job runtime 40% via AQE, broadcast join optimization, skew-key salting, JVM tuning, and memory tuning — reduced AWS EMR cluster cost ~30% for the same throughput.
- Built Delta Lake Lakehouse (partition writes, Z-ordering, OPTIMIZE/VACUUM) — 3x query throughput vs. raw Parquet; implemented exactly-once semantics with idempotent producers and transactional Delta writes.
- Designed observability framework (freshness, volume, schema drift) across 20+ stages — cut MTTD from 4 hours to 10 minutes; established on-call runbooks reducing MTTR by 35%.
- Led schema evolution with Avro + Schema Registry (full-compatibility mode) — zero-downtime upgrades across 15+ downstream services for 12 months of production operation.
- Mentored 4 engineers via Spark internals study groups and structured code reviews — 2 promoted within 12 months; on-call incidents dropped 40%.

Software Engineer – Data Integration · Sirion · Bengaluru, India

Dec 2021 – Jan 2025

- Designed full OLTP→OLAP migration (MySQL to Redshift) with SCD Type 2 dimensional model — enabled sub-second KPI queries, unblocked product analytics initiative, and delivered complete regulatory audit trail.

- Built custom NiFi processors integrated with Spring Boot microservices for 500K+ contract records/month — 50% reduction in manual data handling. Tuned Redshift physical design with KEY/EVEN distribution, compound sort keys, and ENCODE compression — 45% query time reduction.
- Implemented automated DQ framework with completeness, uniqueness, and referential integrity checks plus schema drift detection and pipeline quarantine — 100% data accuracy SLA across all client compliance audits, 70% drop in data incidents over 3 years.
- AWS Glue Data Catalog coverage reduced analyst onboarding from 2 weeks to 3 days; eliminated 10+ weekly ad-hoc data requests to the engineering team.

Senior Data Integration Developer · SmartServ · Bengaluru, India

Feb 2020 – Dec 2021

- Migrated legacy ETL to NiFi distributed architecture — authored trade-off analysis vs. Kafka Streams, drove engineering consensus; fault-tolerance strategy with retry, backpressure handling, and DLQ achieved 25% maintenance cost reduction and zero data loss across 10+ upstream sources.
- Built data cleansing and enrichment pipelines improving BI accuracy by 25%; Tableau self-service integration eliminated 15+ eng-hours/week of analyst support and onboarded 3 business units to independent reporting.
- Facilitated cross-functional data design workshops — produced formal data contracts and schema agreements between business, product, and engineering, reducing downstream rework by 30%.

SELECTED PROJECTS

SPB Room Availability Pipeline — Marriott International · Spark · Kafka · NiFi · EMR · Delta Lake · Avro

- Architected real-time stateful streaming pipeline processing 10K+ events/sec across 500+ hotels with sub-5s end-to-end SLA. Chose Spark Structured Streaming over Flink after trade-off analysis. Exactly-once semantics via idempotent producers + transactional Delta writes eliminated 2–3% duplicate inventory inflation.
- Delta Lake storage optimized with ZORDER on high-cardinality columns and automated OPTIMIZE + VACUUM schedules — 60% query scan cost reduction vs. raw Parquet. Statistical anomaly detection cut silent failure MTTD from 4 hours to 10 minutes.
- Impact: 40% booking error reduction, multi-million revenue protected at peak, 99.9% production uptime.

CI vs OY Room Block Reconciliation — Marriott International · PySpark · SQL · Delta Lake · S3

- Stateful PySpark engine with checkpointing strategy, incremental processing, and idempotent writes — SLA reduced from 2 days to 2 hours (24x improvement). QA effort down 60%, 2 FTEs freed from manual work. Zero data quality escalations post-launch.

Contract Data Ingestion Platform — Morgan Stanley · NiFi · Kafka · Salesforce API · Redshift

- Multi-modal ingestion at 2M+ docs/month with backpressure handling, DLQ strategy, and autoscaling via partitioned Kafka topics — 30% reliability gain, zero message loss under upstream API outages.
- Redshift schema tuning (distribution keys, compound sort keys, ENCODE compression) — 45% query time reduction. Replaced 48-hour data lag with near-real-time access for 200+ legal and procurement professionals.

Reusable NiFi Processor Framework for Flow Control · Java · Groovy · Spring Boot · Apache NiFi

- Designed a config-driven, environment-aware NiFi processor framework for automated flow lifecycle management — enabled one-click deployment across Dev, Staging, and Production with backpressure handling, self-healing retry, and centralized alerting, reducing manual ops toil by 30%.
- Integrated with Spring Boot microservices via secured REST APIs for cross-environment orchestration — eliminated environment-specific configuration drift across all 3 deployment tiers.

EDUCATION & CERTIFICATIONS

B.Tech, Electronics & Communication Engineering — Vellore Institute of Technology, Vellore 2020 · Score: 83.9% · Top 10% of Batch

Certifications: Databricks Certified GenAI Engineer Associate · HackerRank Problem-Solving – Intermediate · Confluent Kafka Certification (in progress, 2025)

Open Source & Community: Active contributor on GeeksForGeeks · HackerRank top percentile in Problem Solving · Spark & Kafka stack discussions on community forums